

Delay-Tolerant Meaconing-Attack Inference for Maritime GNSS via Collaborative OSNMA State Analysis

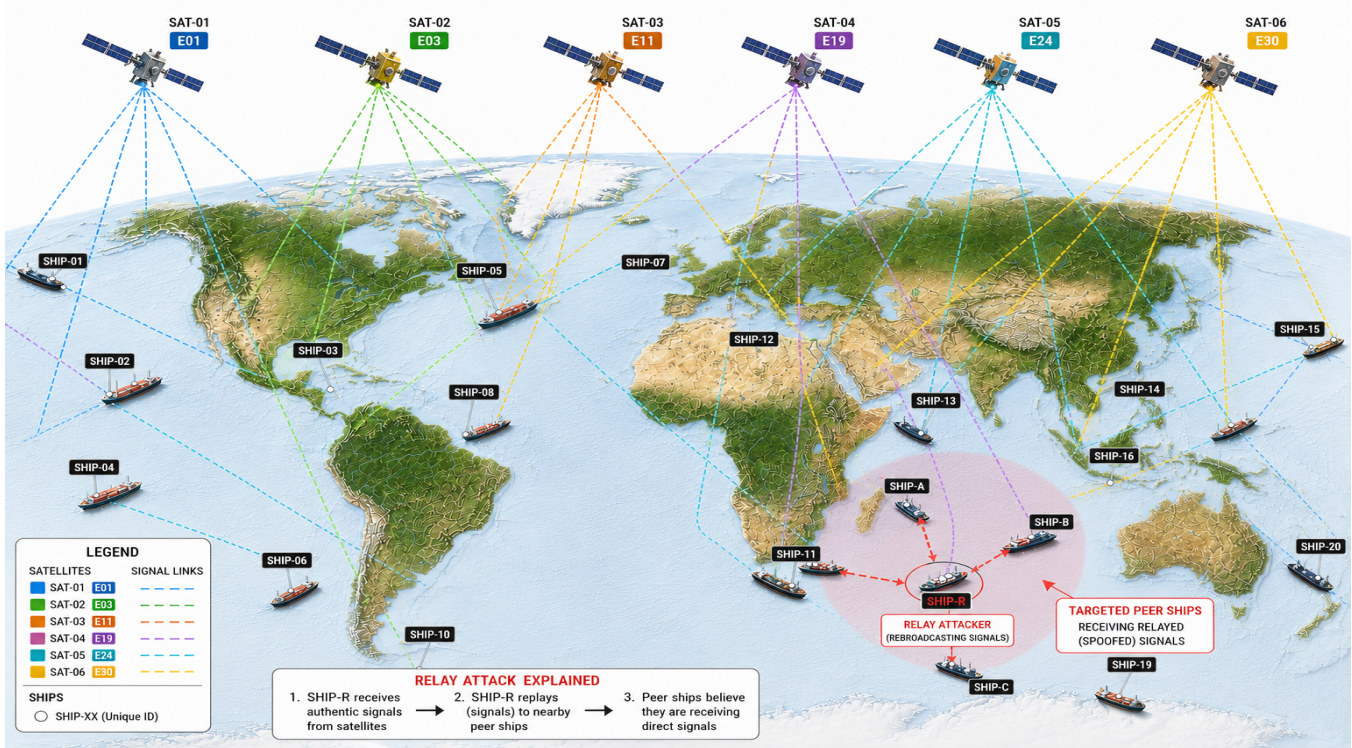


Figure 1: Global view of OSNMA signal reception and meaconing attack: 6 Galileo satellites (E01, E03, E11, E19, E24, E30) broadcast authenticated signals received by multiple ships. In the ocean SHIP-R acts as a meaconing attacker; it receives authenticated signals from the satellite and rebroadcasts them to nearby ships, which may accept the relayed signals.

Abstract

Meaconing/relay attacks — wherein genuine, OSNMA-authenticated Galileo signals are amplified and forwarded from the attacker’s location to a geographically displaced receiver — remain undetected by any single-receiver cryptographic mechanism, including Galileo’s operational Open Service Navigation Message Authentication (OSNMA). Existing multi-receiver defences require physical proximity between receivers or fixed external reference infrastructure, rendering them inapplicable to globally distributed fleets. In this work, we present a novel relay attack detection architecture that exploits a fundamental and previously unexploited property of OSNMA: OSNMA receivers on Earth with an overlapping satellite set

is synchronized to the same TESLA key chain state at any given UTC epoch. Each receiver reports its OSNMA state — chain identifier, disclosed key index, pending MAC tag set, and sub-epoch page counter — signed with a hardware-rooted W3C Decentralized Identifier (DID) which is propagated across fleets. The system detects relay-attacked receivers as TESLA-state outliers against the honest majority.

CCS Concepts

• Computer Networks, Computer Security, Distributed Systems;

Keywords

Galileo satellites, OSNMA, TESLA, DID

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1 Introduction

On July 24, 2025[3], the Galileo Open Service Navigation Message Authentication (OSNMA) service became active, becoming the first civilian GNSS constellation to offer cryptographic authentication of its navigation signals. OSNMA employs the TESLA (Timed Efficient Stream Loss-tolerant Authentication) protocol[4], making signal forgery computationally infeasible for an attacker, as any receiver that implements the full TESLA key chain verification signal authenticity can be easily verified. This represents a decisive advance against the dominant class of GNSS spoofing attacks. However, it is acknowledged — in the OSNMA ICD itself and in the academic literature — to provide no protection against meaconing (relay) attacks[13][7].

A relay attack involves retransmitting genuine signals that are authenticated by OSNMA from an actual location to a receiver located elsewhere. Since the signals are legitimate, the OSNMA verification is successful. Fig. 1 shows a scenario of a relay attack in the ocean. The crisis in the Strait of Hormuz in 2026, where more than 1,100 vessels faced synchronized GPS disruptions within a single day [5], along with the 1,900 recorded GPS interference incidents at eight airports in India from 2023 to 2025 [6], highlight the pressing need to address this vulnerability. Current methods for detecting relay interference are either based on single-receiver strategies (such as clock offset anomaly, Doppler consistency, and IMU divergence [12]) that adept attackers can circumvent, or multi-receiver geometric techniques (like TDOA double differences) that necessitate the physical closeness of receivers. Neither approach is suitable for a globally distributed fleet, such as a naval organization or a shipping company, where receivers are situated across vast ocean regions.

This paper exploits a property of OSNMA that, to our knowledge, has not previously been formalized as a spoofing detection mechanism: under nominal reception conditions, the TESLA key chain state is globally synchronized across honest receivers. Receivers tracking Galileo E1 at the same GST epoch therefore converge on the same disclosed TESLA key index, maintain consistent pending MAC tag sets for commonly observed satellites, and share a common sub-epoch page progression. Persistent deviations from this shared authentication state can therefore serve as indicators of spoofing or protocol manipulation. A relay-attacked receiver, processing signals delayed by the relay path, will deviate from this global state in ways that are precisely quantifiable and detectable at a fleet management server — without any external reference infrastructure.

2 Background

2.1 OSNMA and the TESLA protocol

OSNMA authenticates Galileo I/NAV navigation messages using the TESLA protocol [4]. A one-way key chain $K[0], K[1], \dots, K[N]$ is generated such that $K[i] = H(K[i+1])$, where H is SHA-256 truncated to 128 bits. Each key $K[i]$ is used to generate HMAC authentication tags for navigation messages transmitted during epoch i and is disclosed two epochs later (a 60-second delay, as specified in ICD v1.1 [4]). Receivers verify the integrity of the disclosed keys by checking the one-way chain relation $K[i] = H(K[i+1])$, anchored by the ECDSA-P256-signed root key (KROOT) distributed by EUSPA.

Galileo I/NAV messages are transmitted in 30-second subframes, each comprising fifteen 2-second pages. Because the TESLA key schedule is defined globally, honest receivers observing the same Galileo System Time (GST) epoch are expected, under nominal reception conditions, to validate the same disclosed TESLA key and accumulate authentication tags generated with the same current key for commonly observed satellites. This shared authentication state forms the basis of the detection approach proposed in this work.

3 Related Works

Single-receiver techniques include C/N0 monitoring[12], AGC level checks[12], clock drift analysis [8], Doppler consistency[14], and IMU/GNSS divergence[15]. Each provides a weak signal that sophisticated relay operators can defeat by careful relay engineering. Adtran Oscilloquartz [15] compares GNSS timing against PTP references — effective but requires PTP infrastructure at every monitoring point. Multi-receiver geometric approaches [9, 10] exploit pseudorange double differences and TDOA inconsistencies but require shared satellite visibility (receivers within 2,000 km). Rustamov et al.[10] demonstrated cooperative anti-spoofing via raw measurement exchange but did not address global fleet deployments or OSNMA state comparison. In [12], a software-only anti-replay detector based on within-symbol partial correlations for Galileo E1-B is introduced. The approach is on-chip, local, and does not require any external communication or information sharing. However, it does not help identify the locality of the attack, affected vessels, nor probabilistic reasoning, or confidence estimation. A comparison table of different approaches proposed for detecting replay attacks is provided in Table 1. Considering [12] as the foundation in this work, we attempt to address its limitations. We have not found any prior works that use the global TESLA key chain state as a detection primitive across geographically separated receivers.

4 System Overview

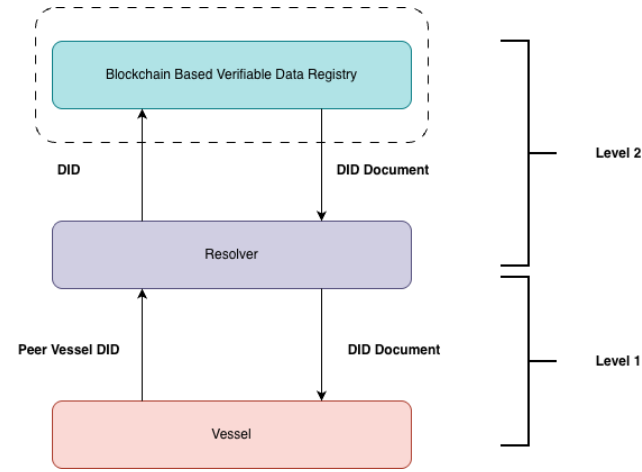
4.1 Vessel Capabilities

Each vessel V_i in a fleet contains the following trusted components:

- (1) *OSNMA Receiver Module*: It receives and processes Galileo E1-B signals, verifies TESLA authentication chains, and maintains local authentication-state evolution.
- (2) *Secure Telemetry Module*: Signs telemetry, also known as a signed observation block (SOB) using a vessel-bound decentralised identity (DID) and transmits authenticated fleet observations.
- (3) *The local integrity monitor*: computes vessel-local anomaly indicators, maintains synchronization statistics, and estimates relay probability.
- (4) *Communication Interface*: exchanges telemetry with fleet infrastructure. SOBs can be transferred over maritime networks—including VDES, AIS extensions, Wi-Fi mesh, LTE/5G, satellite links, and delay-tolerant opportunistic networks—while remaining resilient to packet loss, communication delays, network partitions, and the absence of shore-side infrastructure. This enables robust GNSS integrity monitoring in both connected and infrastructure-denied maritime environments.

Table 1: Comparison of OSNMA-Based Spoofing and Replay Detection Mechanisms

Work	Year	OSNMA-Based	Detection Principle	Replay Detection	Infrastructure Required	Distributed Receivers	Delay-Tolerant
O'Driscoll et al. [9]	2020	Yes	Symbol unpredictability verification	Partial	No	No	Yes
Seco-Granados et al. [11]	2021	Yes	Partial correlation of unpredictable symbols	Yes	No	No	Yes
Arizabaleta-Diez et al. [1]	2024	Yes	OSNMA with antenna-array angle-of-arrival analysis	Partial	Multi-antenna array	No	No
Wang et al. [14]	2025	Yes	Practical replay and forgery attacks against OSNMA	No	SDR hardware	No	Yes
Suarez et. al. [12]	2026	Yes	Practical replay and forgery attacks against OSNMA	No	No	No	No
(This Work)	2026	Yes	Vessel-satellite constraint propagation over global authentication-state graph	Yes	No	Yes	Yes

**Figure 2: DID Resolution for a Vessel DID**

```

{
  "id": "did:vessel:0xb96ab6f1A9a20276512F731A455Ed21784eEcb5c",
  "controller": "did:vessel:0x14149d43c82f08a3bb5a07586A00a55126A41107",
  "vesselName": "Zubeen",
  "operatedBy": "NavShip Ocean",
  "nationality": "Indian",
  "verificationMethod": [
    {
      "id": "did:vessel:0xb96ab6f1A9a20276512F731A455Ed21784eEcb5c#key-ais-1",
      "type": "JsonWebKey2020",
      "publicKeyJwk": { "kty": "EC", "crv": "P-256" }
    }
  ],
  "service": [
    { "type": "FlagStateRegistry", "endpoint": "did:vessel:flagstate:MI" },
    { "type": "ClassCertificate", "endpoint": "did:vessel:classociety:dnv" },
    { "type": "RevocationStatus", "endpoint": "https://registry.vessel.eu/..." }
  ]
}

```

Figure 3: Vessel DID

4.2 OSNMA Authentication-State Model

Let $S_i(t, s)$ denote the authentication state observed by vessel V_i for the satellite s at Galileo System Time t .

The authentication state includes:

- (1) authenticated page counters
- (2) TESLA key disclosure progression
- (3) tag verification status
- (4) authentication timing metadata

We define the visible satellite set for a vessel V_i at time t as the following:

$$G_i(t) = \{s_1, s_2, \dots, s_k\}$$

Importantly, the system does not assume globally identical satellite visibility. Instead, collaborative comparison is restricted to overlapping visibility subsets:

$$G_{i,j}(t) = G_i(t) \cap G_j(t) \text{ for vessel pairs } V_i, V_j$$

5 Authentication Layer

While Galileo Open Service Navigation Message Authentication (OSNMA) guarantees the authenticity and integrity of navigation messages transmitted by Galileo satellites, it does not authenticate the identity of vessels exchanging observations. To establish trust among collaborating vessels we need a robust, decentralised authentication mechanism with high availability. Decentralised Identity (DID) turns out to be a potential player.

A DID is a globally unique cryptographic identifier anchored preferably on a distributed ledger such as a blockchain. For a vessel, it is issued by the flag state authority at registration — analogous to the MMSI, but unlike the MMSI, it is cryptographically bound to a key pair stored on the vessel. The vessel's DID document, as shown in Fig. 3, is publicly resolvable by anyone on the ledger and contains the vessel's public key, vessel name, nationality, revocation endpoint etc. It is a machine-verifiable identity card for the vessel.

A DID-based approach provides several security benefits:-

- (1) Authentication: Confirms the identity of the vessel producing each observation.
- (2) Authorisation: Ensures that only credentialed vessels participate in collaborative inference.
- (3) Non-Repudiation: Cryptographically binds every transmitted observation to its originating vessel.

The vessel DID is bound to the vessel's hardware, which makes it immune to identity theft, as the only compromise of the keys will not allow the attacker to sign messages on behalf of the vessel. A successful attack would rather ask for compromise of the vessel's hardware for which physical access to the vessel is needed, which is much more difficult.

5.1 Vessel DID Registration/Revocation

A new vessel joining the fleet generates its public key and private key pair on its own. The vessel also forms its DID from the generated public key. This DID uniquely identifies the ship in the ocean. Whereas, the private key is used for signing the transmitted data. A competent authority such as the shipping company is responsible for the registration of the DID of the concerned vessel as shown in Fig. 3. The authority forms a DID document containing characteristics of the vessel. The DID Document is bound to the DID of the vessel. The registration is done on some trusted shared platform such as the blockchain. Similarly, the concerned shipping company is entrusted for vessel DID revocation.

5.2 Vessel DID Resolution

Vessel DID needs to be resolved to obtain information specific to the vessel such as the nationality of the vessel in the international waters. A vessel receiving a vessel DID from its peers resolves the DID.

We introduce a two-layer approach for DID resolution, as shown in Fig. 2. At Layer 1, vessels maintain a local datastore of known vessel DID's and their corresponding DID document. When a vessel DID arrives it is searched in the local datastore with the help of a resolver. If no entry is found in the local datastore a query is made to the public Verifiable Data Registry (VDR). This comes handy when the vessel is operating in the middle of the ocean where proper communication infrastructure is unavailable. Vessels can sync periodically or opportunistically to the Layer 2 VDR to update the state of local datastore.

6 Proposed Algorithm

The algorithm, Algorithm 1 is a collaborative integrity-checking process for a fleet of vessels. Each vessel shares signed observation blocks (SOBs) containing what satellites it sees and integrity-relevant fields (TESLA key index, page counter, MAC digest, Doppler sketch, timestamp).

Listing 1: Structure of a SOB

```

1 {
2   "vessel_id": "vessel-0015",
3   "event_time": "2023-08-16T05:00:01Z",
4   "osnma_key_chain_context": {
5     "chain_id": "CHAIN-01",
6     "pkid": 1,

```

```

7   "tesla_root_key_id": "KROOT-2023-08",
8   "tesla_root_key_hash": "4
9     E7A1D9BC2F355A08D91EE2A7C4B19DF",
10   "key_epoch": {
11     "gst_week": 1251,
12     "gst_tow_seconds": 277201
13   },
14   "did_identity": "did:osnma:vessel-0015",
15   "signature": "0x7448cb34...876a9f",
16   "observations": [
17     {
18       "satellite_id": "E01",
19       "tesla_key_index": 13,
20       "page_counter": 1566,
21       "mac_tag_digest": "A5F23E7C00",
22       "doppler_hz": 1035.0,
23       "observation_time": "2023-08-16T05:00:
24         01Z"
25     },
26     {
27       "satellite_id": "E02",
28       "tesla_key_index": 13,
29       "page_counter": 1656,
30       "mac_tag_digest": "A5F23E7C11",
31       "doppler_hz": 644.0,
32       "observation_time": "2023-08-16T05:00:
33         01Z"
34     }
35   ]
36 }

```

The system first filters out any SOB that fails basic trust checks (invalid signature/identity, expired, replayed). It then keeps only recent SOBs in a rolling event-time window and builds a bipartite graph that links vessel nodes to satellite nodes through authenticated observation edges.

Next, for each satellite, the algorithm estimates how internally consistent the fleet's observations are by checking temporal continuity, key evolution, page progression, MAC agreement, and Doppler smoothness. Using this satellite-level consistency as reference, it computes per-edge vessel-satellite residuals (how much each observation deviates). These local residuals are then propagated through shared vessel-satellite paths so each vessel's score reflects both its own behavior and nearby collaborative evidence. After that, the algorithm computes a manifold-style global residual per vessel (how far that vessel sits from the expected joint consistency structure) and a complementary consistency score.

Finally, it fuses this evidence into a vessel-level divergence probability $P_i^{div}(t)$. The fusion is trust-weighted by observation freshness using an exponential decay term $\omega_i = e^{-\lambda \Delta t_i}$, and combined with a prior via Bayesian-style posterior update. The output is one divergence estimate per vessel: higher values mean the vessel is more likely to be compromised or spoofed.

7 Experiment and Results

7.1 Experimental Setup

We evaluate collaborative OSNMA integrity inference using a protocol-driven pipeline that fuses official Galileo OSNMA test vectors[2], Orekit[8]-derived visibility geometry, and synthetic maritime fleet observations. The objective is vessel-level spoofing detection via SCP-derived divergence posterior (P_{div}).

7.2 Dataset

The OSNMA input follows the official test-vector[2] structure and is parsed from CSV records with fields SVID, NumNavBits, and NavBitsHEX. Filenames encode GST snapshot timestamps. The packaged bundle currently contains 18 unique GST timestamps across 18 scenarios.

7.3 Overlap Statistics:

The geometry input contains receiver-indexed visibility observations with fields $receiver_id$, epoch, SVID, $elevation_{deg}$, $azimuth_{deg}$, and $range_km$. In the evaluated geometry file, we observe:

30 total observations 14 unique vessels ($receiver_id$) 3 unique epochs 5 unique satellites

4.2 Overlap Statistics Because SCP relies on shared satellite constraints, overlap quality is a key experimental condition. For the current geometry file:

- (1) 6 unique slots
- (2) 6 overlap events with vessels observing the same satellite at the same epoch
- (3) 32 vessel pairs with at least one shared observation

Highest overlap occurs at epoch 2026-06-12T00:00:00Z, where satellites 2, 3, and 7 are each observed by 7 vessels.

Vessels are placed around three fixed geographic centers and then jittered locally. Fleet sizes of 120 and 200 vessels are evaluated across seeds 11 and 12, with spoofing injected into 2% and 5% of vessels, and attack targets constrained to vessel IDs that are actually observed in the geometry file.

7.4 Results

We evaluate our work on official Galileo OSNMA test vectors with Orekit-derived visibility geometry, using a grid over seeds 11, 12, fleet sizes 120, 200, and attack fractions 2%, 5%. Thresholds were swept on a log scale from 0.0005 to 0.2. The result metrics are shown in the table 2:-

Table 2: Detection Performance

Metric	Value
ROC-AUC	0.7901
PR-AUC	0.8538
AP	0.8715
F1	0.8987
Precision	0.8274
Recall	1.0000

These values indicate high separability in both ROC and precision-recall space, with perfect recall at the selected operating point and

strong precision, supporting effective attack identification in the evaluated overlap-aware setting.

A key experimental control was restricting attack assignment to geometry-observed vessels and using clustered fleet layouts to ensure meaningful cross-vessel overlap. Under this observability-aligned protocol, detection performance improved substantially compared with earlier misaligned sampling runs (near-random behavior). This confirms that overlap and observability are first-order determinants of collaborative OSNMA inference performance, and should be treated as required evaluation constraints rather than optional scenario details.

8 Conclusion

We presented a relay attack detection system for globally distributed GNSS maritime fleets that requires limited external reference infrastructure. Our work exploits the global synchronisation of the OSNMA TESLA key chain as a relay timing anomaly detector: a relay-attacked receiver's TESLA state is detectably behind the honest fleet majority, and its DID-signed SOB makes this evidence non-repudiable.

Unlike existing OSNMA-based defences that fully rely on single-receiver analysis, synchronised nearby receivers, or specialised RF infrastructure, the proposed framework exploits globally shared TESLA authentication-state evolution across independent receivers. By modeling collaborative observations as a vessel-satellite integrity graph and propagating consistency constraints through overlapping satellite visibility, The work enables replay-induced anomalies to be identified even when spoofed vessel populations are internally consistent and numerically dominant. Vessels opportunistically exchange compact signed Security Observation Bundles (SOBs) containing authenticated OSNMA state summaries, enabling collaborative integrity inference without requiring dedicated monitoring infrastructure or trusted terrestrial reference stations. Through graph-consistent satellite-state estimation, temporal consistency analysis, and trust-weighted evidence fusion, replay detection is transformed from a purely local RF problem into a distributed cryptographic integrity inference problem.

Preliminary results demonstrate that even partial satellite overlap between geographically separated honest and spoofed vessel populations is sufficient to expose globally inconsistent authentication-state trajectories introduced by replay attacks. Furthermore, receiver-level replay detectors (such as the partial-correlation SCER detector) could be incorporated as an additional evidence source in the Satellite Observation Block and fused within the Bayesian inference framework. Overall, the study supports the feasibility of collaborative OSNMA-based spoofing detection in fleet environments and establishes a reproducible evaluation pipeline for protocol-centric validation.

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Algorithm 1: Collaborative OSNMA State Analysis

Input: Asynchronously received signed SOB stream $\mathcal{S}$
Output: Integrity-divergence estimates $\{P_i^{div}(t)\}$ for all vessels

1. SOB Verification and Event-Time Buffering

foreach *received SOB* $s_i \in \mathcal{S}$ **do**
 verify DID identity and digital signature
 reject invalid, expired, or replayed SOBs
 insert verified SOB into rolling event-time cache

2. Event-Time Window Construction

construct rolling event-time window

$$W_t = [t - \Delta, t]$$

retrieve all SOBs whose event timestamps belong to W_t

3. Bipartite Integrity Graph Construction

construct bipartite graph

$$\mathcal{G}(t) = (\mathcal{V} \cup \mathcal{S}, \mathcal{E})$$

where:

- $\mathcal{V}$ represents vessel nodes
- $\mathcal{S}$ represents satellite-state nodes
- $\mathcal{E}$ represents authenticated observation edges

foreach *vessel observation* s_i **in** W_t **do**

foreach *visible satellite* $k \in G_i$ **do**

 create edge

$$(V_i, S_k) \in \mathcal{E}$$

 attach edge attributes:

- TESLA key index
- authenticated page counter
- MAC-tag digest
- Doppler sketch
- observation timestamp

4. Satellite-State Consistency Estimation

foreach *satellite node* S_k **do**

 estimate globally consistent authentication-state trajectory using:

- temporal continuity
- TESLA key evolution
- authenticated page progression
- MAC-tag agreement
- Doppler smoothness

5. Constraint Propagation Across Fleet Graph

propagate integrity constraints through shared satellite observations

foreach *edge* (V_i, S_k) **do**

 compute local consistency residual

$$R_{ik} = \text{divergence}(V_i, S_k)$$

 propagate residuals across neighboring vessel-satellite paths

6. Global Manifold Embedding Analysis

foreach *vessel* V_i **do**

 compute global embedding residual

$$E_i = \text{manifold_deviation}(V_i)$$

$$C_i = \text{consistency}(V_i)$$

7. Temporal Trust-Weighted Evidence Fusion

foreach *vessel* V_i **do**

$$\omega_i = e^{-\lambda \Delta t_i}$$

$$P_i^{div}(t) \propto P(E_i, C_i \mid \text{divergence}) \cdot P_i^{\text{prior}}$$

return $\{P_i^{div}(t)\}$

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